09 IMMORTALITY

We are on the verge of many life extension techniques that could allow humans to practically live forever. For some scientists, senescence or the deterioration of the body and mind in old age is just another disease to be cured. At the very least, life spans are expected to increase If not physical immortality, we may at least be able to upload our brains to machines.

10 LEVEL 5 AUTONOMOUS VEHICLES

We are currently at Level 2 autonomous cars, where the driver still has to continuously monitor the vehicle. Concept vehicles are at level 4, where a drive can intervene when needed. Level 5 autonomous cars require no input from the rider except setting a destination. The entire automotive industry is moving towards level 5 autonomy, and we can expect such vehicles to hit the road within the next decade.

11 MALE BIRTH CONTROL

This breakthrough is expected to hit the market in the next five years, as it has already proven effective in clinical trials. Within our lifetimes, we will see pills that males can consume to prevent pregnancy, they have minimal side effects, and no permanent effect on bodies. The introduction to the market will be a culmination of decades worth of research.

12 MANNED MISSION TO MARS

NASA, ESA, CNSA and private space companies such as SpaceX, Boeing, and Blue Origin are working on the rockets, landers, vehicles and other equipment necessary to support the first manned mission to Mars. There are at least five active planned missions before the year 2060. While the funding is not complete yet, NASA has a timeline for a manned mission by 2033.

13 MINING THE MOON

The Moon has buried reserves of water and helium, both of which can turn out to be immensely useful for future space missions, agriculture and fuel production. The real gold rush though, for private companies at least, would be for the rare earth metals, which are essential for the production of electronics The moon is abundant on these metals, known as lanthanides.

14 NANOBOTS FIGHT CANCER

Cancer is caused by any kind of stress on tissues, and involves cells in the body going rogue and reproducing at a rapid rate. Therefore, it has proved incredibly difficult to cure cancer. However, a new form of therapy involving nanobots in the body that fight the cancerous tissue on a cellular level, could drastically improve various therapies. Researchers have already developed autonomous DNA robots to fight cancer.

15 SPACE ELEVATOR

We are finally engineering materials with the weight and strength required to construct orbital towers, or elevators to transport people and things from the surface of the Earth to outer space. This is most likely going to be graphene or a similar material. Such a space elevator would have to be built at incredible cost, but would lower the cost of access to space. The concept is explored in the novel, The Fountains of Paradise.

16 SPACE TOURISM

While some individuals have already financed their own spaceflights, space tourism is expected to get more common and cheaper in the future. While most of the concepts are focused on hotels in low earth orbit, we might see at least flights to the Moon. SpaceX already has plans to send about eight people on a round trip to the Moon.

17 TELEPATHY

If the human connectome project is a success, and there are reliable machine to brain interfaces developed, then the internet could allow human brains to communicate directly with each other, realising a technological form of telepathy. Such a development would be a paradigm shift in the way humans function, taking us closer to the technological singularity, and the emergence of a global human hivemind.

Uber "Skyport" of Corgan design

18 THE DISCOVERY OF ET

The hunt for life will heat up with the launch of astronomical instruments that can peer into the surfaces of exoplanets. A number of missions have been undertaken to explore the moons of the gas giants. We might find signs of life on Europa, Titan, Ganymede, or in one of the seven promising planets in the Trappist-1 system. Another possibility is finding the traces of microorganisms on one of the future Mars missions.